

File permissions in Linux

Project description

The research team at my organization needs to update the file permissions for certain files and directories within the projects directory. The current permissions do not fully reflect the level of authorization that should be given. Checking and updating these permissions helps keep the organization's files and directories secure. To complete this task, I performed the following tasks:

Check file and directory details

The following code demonstrates how I used Linux commands to determine the existing permissions set for the files and directories in the file system.

The first command in the screenshot uses `ls -la` to list all contents of the projects directory. I used the `-l` option to display detailed information about each file and directory and the `-a` option to include hidden files. The output shows one directory named `drafts`, one hidden file named `.project_x.txt`, and four other project files. The first column of the output contains a 10-character permissions string that shows the type of file and the permissions assigned to the user, group, and other users.

```
researcher2@fa8a3eae8d97:~/projects$ ls -la
total 20
drwx--x--- 2 researcher2 research_team 4096 Sep  5 17:43 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Sep  5 17:43 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep  5 17:43 project_m.txt
-rw-rw-r--  1 researcher2 research_team  46 Sep  5 17:43 project_r.txt
-rw-rw-r--  1 researcher2 research_team  46 Sep  5 17:43 project_t.txt
researcher2@fa8a3eae8d97:~/projects$
```

Describe the permissions string

The 10-character permissions string can be deconstructed to determine who is authorized to access a file or directory and which permissions they have. The characters and what they represent are as follows:

- 1st character: This character indicates the file type. If it is `d`, the item is a directory. If it is a hyphen (`-`), it is a regular file.

- 2nd–4th characters: These characters represent the user permissions: read (r), write (w), and execute (x). A hyphen (-) means that the corresponding permission is not granted.
- 5th–7th characters: These characters represent the group permissions: read (r), write (w), and execute (x). A hyphen means that the corresponding permission is not granted to the group.
- 8th–10th characters: These characters represent the permissions for other users. Other users are users on the system who are not the file owner and are not members of the file's group. A hyphen means that the corresponding permission is not granted.

For example, the initial permissions for `project_k.txt` are `-rw-rw-rw-`. Since the first character is a hyphen (-), this indicates that `project_k.txt` is a regular file. The user has read and write permissions, the group has read and write permissions, and other users also have read and write permissions. No users have execute permissions.

The permissions for the `drafts` directory are `drwx--x---`. Since the first character is `d`, this indicates that `drafts` is a directory. The user has read, write, and execute permissions. The group has execute permission, while other users have no permissions.

Change file permissions

The organization determined that other users should not have write access to any of the project files. To comply with this requirement, I reviewed the existing permissions and determined that `project_k.txt` had write permission granted to other users.

The following code demonstrates how I used Linux commands to change the permissions:

The first command in the screenshot is `chmod o-w project_k.txt`. The `chmod` command is used to change the permissions of a file or directory. The `o` represents other users, and `-w` means to remove write permission. Therefore, this command removes write permission from other users for `project_k.txt`.

After making the change, I used `ls -la` to review the updated permissions. The permissions for `project_k.txt` changed from `-rw-rw-rw-` to `-rw-rw-r--`. This means that the user and group can still read and write to the file, while other users can only read it.

```
researcher2@fa8a3eae8d97:~/projects$ chmod o-w project_k.txt
researcher2@fa8a3eae8d97:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 17:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 18:16 ..
-rw--w---- 1 researcher2 research_team  46 Sep  5 17:43 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep  5 17:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep  5 17:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_t.txt
researcher2@fa8a3eae8d97:~/projects$
```

Change file permissions on a hidden file

```
researcher2@fa8a3eae8d97:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 17:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 18:16 ..
-rw--w---- 1 researcher2 research_team  46 Sep  5 17:43 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep  5 17:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep  5 17:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_t.txt

researcher2@fa8a3eae8d97:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@fa8a3eae8d97:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 17:43 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep  5 18:16 ..
-r--r----- 1 researcher2 research_team  46 Sep  5 17:43 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Sep  5 17:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep  5 17:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_t.txt
researcher2@fa8a3eae8d97:~/projects$
```

The research team also needed to restrict write access to the hidden file `.project_x.txt`. The file is hidden because its name begins with a period (`.`).

The original permissions for `.project_x.txt` were `-rw--w----`. This means that the user had read and write permissions, the group had write permission, and other users had no permissions.

The following code demonstrates how I used a Linux command to change the permissions:

I used `chmod u-w,g-w,g+r .project_x.txt` to update the permissions. The `u` represents the user, and `-w` removes write permission from the user. The `g` represents the group. The `g-w` removes write permission from the group, while `g+r` adds read permission for the group.

After making these changes, I used `ls -la` to verify the updated permissions.

The permissions changed from `-rw--w----` to `-r--r-----`. This means that the user has read permission but no longer has write permission, the group has read permission but no longer has write permission, and other users have no permissions.

Change directory permissions

My organization only wants the `researcher2` user to have execute permissions for the `drafts` directory. This means that the group should not have execute permission, and other users should not have access to the directory.

The following code demonstrates how I used Linux commands to change the directory permissions:

I first used `ls -l` to view the permissions of the files and the `drafts` directory. The permissions for `drafts` were `drwx--x---`. The first character, `d`, indicates that `drafts` is a directory. The user already had read, write, and execute permissions, while the group had execute permission.

I then used the command `chmod g-x drafts/`. The `g` represents the group, and `-x` removes execute permission from the group. The user already had execute permission, so no change was needed for the user.

After running the command, I used `ls -l` again to verify the changes. The permissions for the `drafts` directory changed from `drwx--x---` to `drwx-----`. This means that only the `researcher2` user has read, write, and execute permissions for the directory. The group and other users no longer have permissions.

```
researcher2@fa8a3eae8d97:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Sep  5 17:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep  5 17:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_t.txt
researcher2@fa8a3eae8d97:~/projects$ chmod g-x drafts/
researcher2@fa8a3eae8d97:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Sep  5 17:43 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep  5 17:43 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep  5 17:43 project_t.txt
researcher2@fa8a3eae8d97:~/projects$
```

Summary

I changed multiple permissions to match the level of authorization required by my organization for files and directories in the projects directory. I first used `ls -la` and `ls -l` to check the existing permissions. This allowed me to identify which files and directories had permissions that needed to be changed. I then used the `chmod` command to remove unnecessary permissions from `project_k.txt`, `project_x.txt`, and the `drafts` directory. Finally, I used the `ls` commands again to verify that the permission changes were applied successfully. These changes help restrict unauthorized access and improve the security of the organization's files and directories.